Green University of Bangladesh
Department of Computer Science and Engineering (CSE)
Faculty of Sciences and Engineering
Semester: (Fall, Year:2024), B.Sc. in CSE (Day)

Lab Report NO #02
Course Title: Database Lab
Course Code: CSE210
Section: D3

Lab Experiment Name: SQL Database Creation and Query Filtering (Library System)

Student Details

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Lab Date : **09/04/26**
Submission Date : **13/04/26**
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Lab Report Status
Marks: ………………………………… **Signature:.....................**
Comments:... **Date:..............................**

1. TITLE OF THE LAB REPORT EXPERIMENT

In this lab experiment, a database named **library_query_lab** was created. Three tables—MEMBER, BOOK, and BORROW—were designed and implemented. Various SQL queries were executed to perform data filtering, sorting, grouping, and aggregation operations.

2. OBJECTIVES/AIM [2 marks]

- To create and use a database
- To design tables with constraints (Primary Key, Foreign Key, NOT NULL, CHECK, UNIQUE)
- To insert data into tables
- To perform data filtering using SQL queries
- To apply sorting, grouping, and aggregate functions

3. PROCEDURE / ANALYSIS / DESIGN [3 marks]

- Create a database
- Design tables:
  - MEMBER
  - BOOK
  - BORROW
- Apply necessary constraints
- Insert sample data
- Execute SQL queries for filtering and analysis

```sql
CREATE DATABASE library_query_lab;
USE library_query_lab;

-- MEMBER TABLE
CREATE TABLE MEMBER (
   MemberID INT PRIMARY KEY AUTO_INCREMENT,
   FullName VARCHAR(100) NOT NULL,
   Email VARCHAR(100) NOT NULL UNIQUE,
   Age INT CHECK (Age >= 12),
   Phone VARCHAR(15) NOT NULL DEFAULT 'N/A'
);
```

```sql
-- BOOK TABLE
CREATE TABLE BOOK (
    BookID INT PRIMARY KEY AUTO_INCREMENT,
    Title VARCHAR(255) NOT NULL,
    ISBN VARCHAR(50) NOT NULL UNIQUE,
    Price DECIMAL(8,2) CHECK (Price >= 0),
    Availability ENUM('AVAILABLE','BORROWED') DEFAULT 'AVAILABLE'
);

-- BORROW TABLE
CREATE TABLE BORROW (
    BorrowID INT PRIMARY KEY AUTO_INCREMENT,
    MemberID INT,
    BookID INT,
    BorrowDate DATE NOT NULL,
    FOREIGN KEY (MemberID) REFERENCES MEMBER(MemberID),
    FOREIGN KEY (BookID) REFERENCES BOOK(BookID)
);
```

4. IMPLEMENTATION [3 marks]

```sql
-- MEMBER
INSERT INTO MEMBER (FullName, Email, Age, Phone) VALUES
('Amin Khan','amin@gmail.com',22,'01711111111'),
('Rahim Uddin','rahim@gmail.com',25,'01811111111'),
('Karim Hasan','karim@gmail.com',17,'01911111111'),
('Anika Rahman','anika@gmail.com',20,'N/A'),
('Sajid Khan','sajid@gmail.com',30,'01611111111'),
('Tanvir Hasan','tanvir@gmail.com',15,'N/A'),
('Arman Hossain','arman@gmail.com',28,'01511111111'),
('Jannat Islam','jannat@gmail.com',19,'01722222222');

-- BOOK
INSERT INTO BOOK (Title, ISBN, Price, Availability) VALUES
('Database Systems','ISBN001',1500,'AVAILABLE'),
('Data Structures','ISBN002',1200,'BORROWED'),
('Operating System','ISBN003',2000,'AVAILABLE'),
('Computer Networks','ISBN004',800,'BORROWED'),
('Machine Learning','ISBN005',2500,'AVAILABLE'),
('Data Science','ISBN006',1800,'AVAILABLE'),
('Artificial Intelligence','ISBN007',3000,'BORROWED'),
```

```sql
('Programming in C','ISBN008',600,'AVAILABLE');

-- BORROW
INSERT INTO BORROW (MemberID, BookID, BorrowDate) VALUES
(1,2,'2024-01-01'),
(2,4,'2024-01-02'),
(3,2,'2024-01-03'),
(4,7,'2024-01-04'),
(5,4,'2024-01-05'),
(6,7,'2024-01-06'),
(7,2,'2024-01-07'),
(8,4,'2024-01-08'),
(1,7,'2024-01-09'),
(2,2,'2024-01-10');

-- Show all records
SELECT * FROM MEMBER;
SELECT * FROM BOOK;
SELECT * FROM BORROW;

-- Distinct availability
SELECT DISTINCT Availability FROM BOOK;

-- Members with Age >= 18
SELECT * FROM MEMBER WHERE Age >= 18;

-- Books with price between 500 and 2000
SELECT * FROM BOOK WHERE Price BETWEEN 500 AND 2000;

-- Books with specific availability
SELECT * FROM BOOK WHERE Availability IN ('AVAILABLE','BORROWED');

-- Names starting with A or ending with n
SELECT * FROM MEMBER
WHERE FullName LIKE 'A%' OR FullName LIKE '%n';

-- Phone not 'N/A' and Age > 20
SELECT * FROM MEMBER
WHERE Phone <> 'N/A' AND Age > 20;
```

```
-- Age not between 12 and 17
SELECT * FROM MEMBER
-- Title contains 'Data' or 'Database'
SELECT * FROM BOOK
WHERE Title LIKE '%Data%' OR Title LIKE '%Database%';

-- Top 3 most expensive books
SELECT * FROM BOOK ORDER BY Price DESC LIMIT 3;

-- 2 cheapest books
SELECT * FROM BOOK ORDER BY Price ASC LIMIT 2;

-- Count records
SELECT COUNT(*) FROM MEMBER;
SELECT COUNT(*) FROM BOOK;
SELECT COUNT(*) FROM BORROW;

-- Average, Min, Max price
SELECT AVG(Price), MIN(Price), MAX(Price) FROM BOOK;

-- Books by availability
SELECT Availability, COUNT(*)
FROM BOOK
GROUP BY Availability;

-- Borrow count per member
SELECT MemberID, COUNT(*)
FROM BORROW
GROUP BY MemberID;
```

5. TEST RESULT / OUTPUT [3 marks]

- All queries executed successfully
- Data filtering, sorting, and grouping worked correctly
- No major errors were found

6. ANALYSIS AND DISCUSSION [3 marks]

What went well?

- Table creation and constraints worked properly
- Queries executed efficiently

Trouble spots:

- Understanding foreign key relationships
- Using ENUM type

Most difficult part:

- GROUP BY and aggregate functions

What I learned:

- SQL filtering techniques
- Data integrity using constraints
- Practical database design

7. SUMMARY:

In this lab, a complete library database system was developed. Various SQL operations such as filtering, sorting, and aggregation were performed successfully. This lab helped in understanding real-world database implementation.